Project Plan, Roles, Risks & KPIs

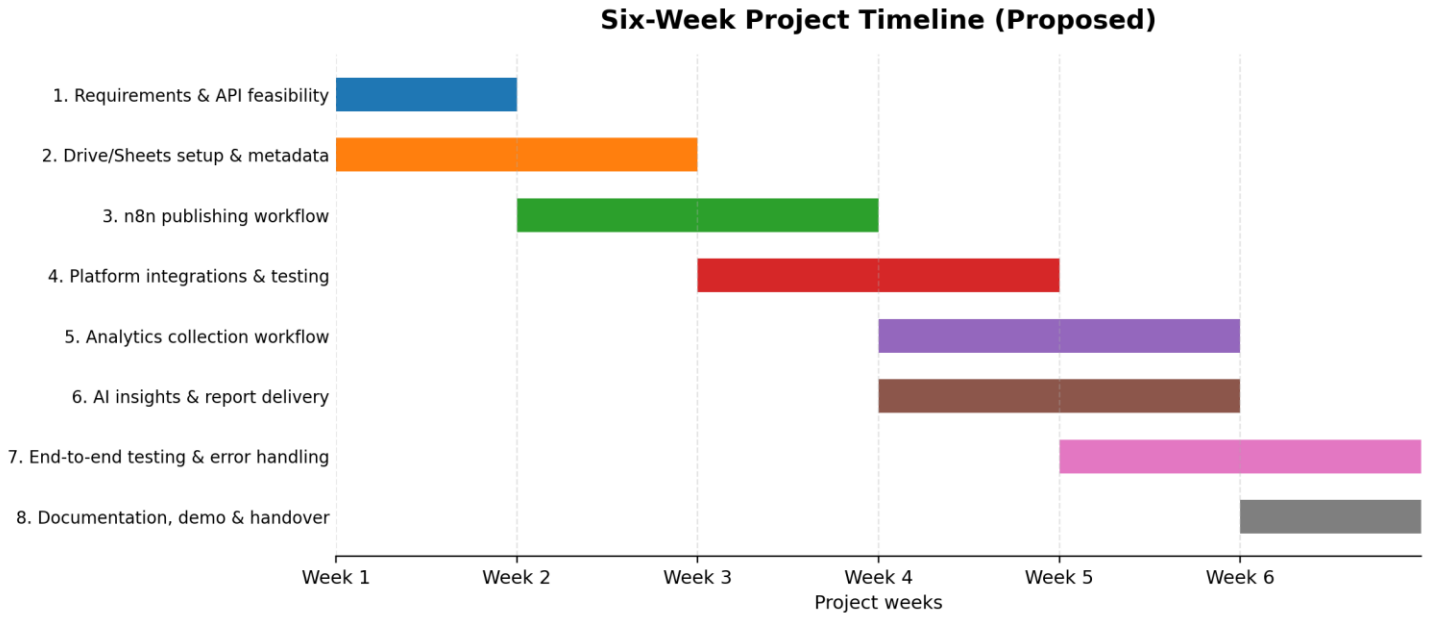
Multi-Platform Shorts Automation & AI Insights Engine

This project plan translates the proposed system workflow into a practical delivery schedule, defined responsibilities, risk controls, and measurable success indicators. The timeline and targets below are proposed planning assumptions and should be confirmed against actual API access and project constraints.

1. Project Plan — Timeline, Milestones, Deliverables, and Resource Allocation

The project is planned over six weeks. Workstreams may overlap where dependencies allow. The plan covers the automated publishing pipeline and the post-publication AI insights feedback loop.

1.1 Proposed Gantt Chart



1.2 Milestones and Deliverables

Milestone	Target	Deliverables / Acceptance Evidence
M1 — Scope & feasibility approved	End of Week 1	Confirmed requirements, API feasibility checklist, credential/access plan
M2 — Data foundation ready	End of Week 2	Google Drive folders, metadata sheet, status/metrics schema
M3 — Publishing workflow prototype	End of Week 3	n8n workflow for trigger, metadata, and initial platform publishing requests
M4 — Platform integrations verified	End of Week 4	Core publishing integrations tested; per-platform status logging verified
M5 — Analytics and AI report ready	End of Week 5	Metrics collection workflow, AI analysis prompt/output, notification and storage path

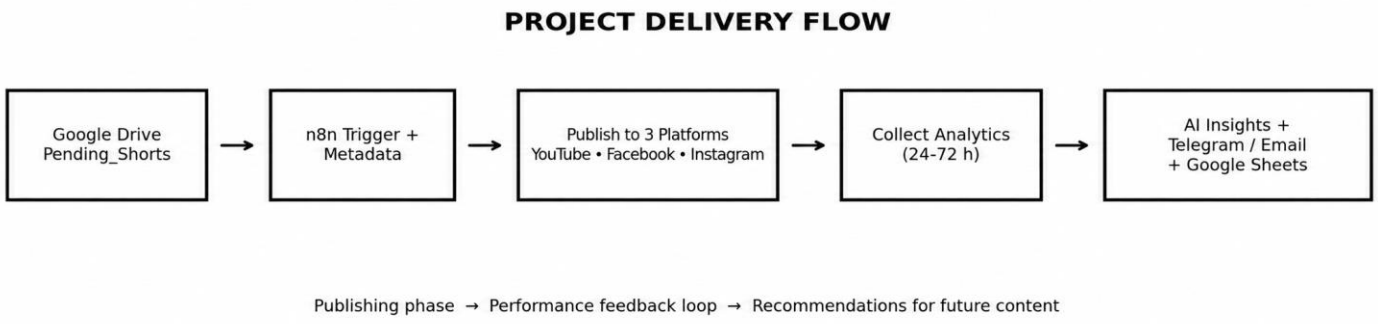
M6 — Tested handover	End of Week 6	End-to-end test results, error-handling checks, workflow documentation, known limitations, handover checklist

1.3 Resource Allocation

Workstream / Resource	Effort Share (Proposed)	Primary Use
Project coordination & requirements	10%	Scope, acceptance criteria, schedule, progress tracking
n8n workflow development	30%	Triggers, routing, metadata, logging, orchestration
Platform API integration	25%	YouTube, Instagram, and Facebook publishing/analytics access
AI analysis & reporting	15%	Prompt design, structured output, report generation
Testing, documentation & handover	20%	Integration tests, failure checks, documentation, demo

Resource note: Percentages are an indicative allocation of total project effort, not confirmed staffing levels. If one
person performs all work, these percentages represent time allocation across responsibilities.

1.4 Project Delivery Diagram



2. Task Assignment & Roles

– Team Leader & Project Manager: leads the team, assigns roles and responsibilities, defines the project milestones and weekly plan, tracks progress against deadlines, and coordinates communication between members.

– API & Integration Lead: researches and obtains the required API keys and credentials (Google, YouTube, Meta, OpenRouter), sets up the business accounts and authorizations, and builds the initial version of the n8n workflow.

– Quality & Optimization Lead: reviews the workflow end to end, tests it, fixes errors, improves performance and reliability, and completes any missing parts or steps in the workflow.

– Documentation & Delivery Lead: prepares the required documents, reports and presentations, and maintains backup copies of the accounts and exported workflows to ensure the project can be recovered and delivered safely.

3. Risk Assessment & Mitigation Plan

Risk	Priority	Potential Impact	Mitigation / Response
Platform API approval or permission limits	High	Publishing or analytics endpoints may require account eligibility, app review, or specific permissions.	Validate API access in Week 1; test one platform first; document unsupported features and fallback options.
Credential expiry or OAuth failure	High	Tokens may expire or permissions may be revoked.	Use secure credential storage; implement token refresh where supported; monitor authentication errors.
Partial publishing failure	High	One platform may reject a request while others succeed.	Track status per platform; log error details; design safe retries and avoid duplicate uploads.
Inconsistent analytics availability	Medium–High	Metrics may differ by platform or become available at different times.	Normalize available fields; record collection timestamps; handle missing metrics explicitly.
Incorrect or unsupported AI recommendations	Medium	The AI may infer patterns from incomplete data or return invalid output.	Use structured output and validation; tie recommendations to observed metrics; label low-confidence conclusions.
Workflow interruption or rate limits	Medium	n8n execution failures or API limits may interrupt processing.	Add error paths, controlled retries, rate-limit handling, and execution monitoring.
Data loss or incorrect file/status mapping	Medium	Video records and metrics may become mismatched.	Use stable video IDs/URLs, validate required fields, and retain execution logs.
Schedule overrun	Medium	API access, integration testing, or debugging may take longer than expected.	Prioritize a minimum viable flow; track weekly milestones; keep buffer time for integration and testing.

3.1 Risk Response Flow

Identify risk	Assess likelihood & impact	Choose mitigation	Monitor trigger or warning	Resolve & document

Risk priority should be reviewed at each milestone. High-priority risks that block a core platform integration should be escalated before expanding the workflow to additional platforms.

4. KPIs — Key Performance Indicators

KPIs measure whether the system is reliable, saves manual effort, collects useful analytics, and delivers actionable insights. Targets below are proposed initial targets for testing; baseline measurements should be collected before final targets are approved.

KPI	Measurement	Proposed Target	Review Frequency	Data Source
Publishing workflow success rate	Successful platform publish attempts ÷ total publish attempts × 100	≥ 95% in controlled tests	Per run / weekly	n8n execution logs + platform responses
Publishing processing time	Elapsed time from file detection to completion of platform requests	Workflow processing ≤ 60 seconds for a test video, excluding platform-side processing delays	Per run	n8n timestamps
Manual effort per video	Time spent by a user on routine cross-posting and status logging	≤ 1 minute of routine manual effort per video after setup	Weekly sample	Before/after time study

Analytics collection success rate	Successful analytics fetches ÷ scheduled fetch attempts × 100	≥ 90% for eligible videos with valid access	Daily / weekly	API response logs
Report delivery success rate	Reports delivered successfully ÷ reports generated × 100	≥ 95%	Weekly	Email/Telegram delivery logs
Data completeness	Required available metric fields populated ÷ required available fields expected × 100	≥ 90%, excluding metrics not exposed by a platform	Weekly	Google Sheets metrics log
Workflow availability	Scheduled monitoring period minus unplanned downtime ÷ scheduled monitoring period × 100	≥ 98% during the pilot, if monitoring is available	Weekly	n8n instance monitoring / execution history
AI report validity	Reports passing schema and required-field validation ÷ reports generated × 100	≥ 95%	Weekly	Output validation logs
Recommendation usefulness	Recommendations rated useful by the project user ÷ recommendations reviewed × 100	≥ 80% in a small pilot review	After each report / monthly	Simple user feedback form

4.1 KPI Measurement Loop

Collect execution & API data	**Calculate KPI values**	**Compare with targets**	**Improve workflow & review again**

5. Acceptance Criteria

A new eligible MP4 file can trigger the configured n8n workflow.

The system attempts publishing through each configured platform integration and records a per-platform outcome.

The workflow stores video metadata and publishing status in Google Sheets and organizes processed files as configured.

The scheduled analytics workflow retrieves available metrics for eligible published videos.

The AI step produces a validated report with findings and actionable recommendations based on available metrics.

The report is delivered through the configured notification channel and the insights are stored for later review.

Known API limitations, failure cases, required credentials, and operating instructions are documented.